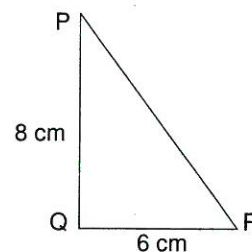
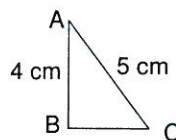


16. Triangle ABC is similar to triangle PQR.
 (a) Complete the following statement and hence find the length of PR.

$$\frac{PR}{5} =$$

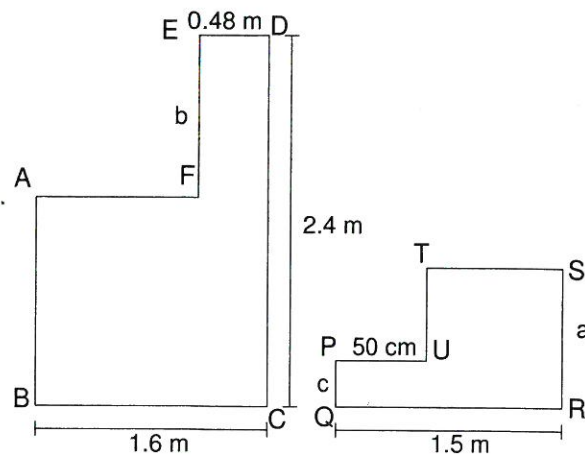


- (b) Find the length of BC.

17. The diagram on the right shows a pair of similar figures.
 (a) Write the similarity statement for these figures.

- (b) Determine the scale factor.

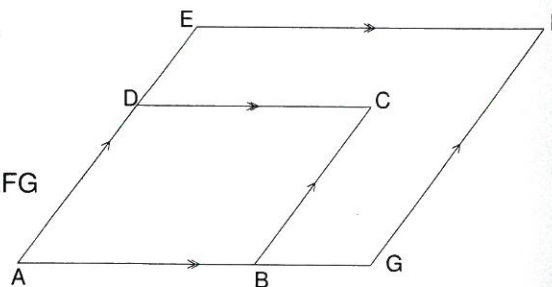
- (c) Calculate the lengths of the sides labelled a, b and c.



18. Parallelogram AGFE has been obtained by enlarging parallelogram ABCD.
 AB = 15.6 cm, AD = 13 cm and AE = 19.5 cm.
 (a) Are these parallelograms similar? Justify your response.

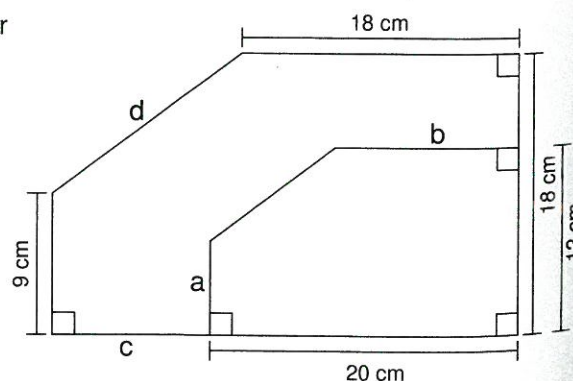
- (b) Calculate the scale factor used to draw parallelogram ACFG the image of parallelogram ABCD.

- (c) Calculate the length of BG.

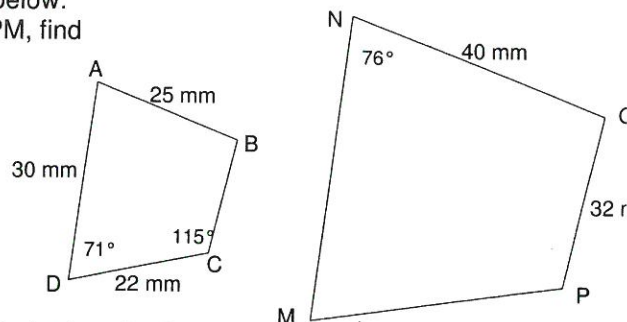


19. The diagram opposite shows two similar pentagons.
 (a) What scale factor has been used to create the smaller pentagon from the larger?

- (b) Calculate the side lengths labelled a, b, c and d.

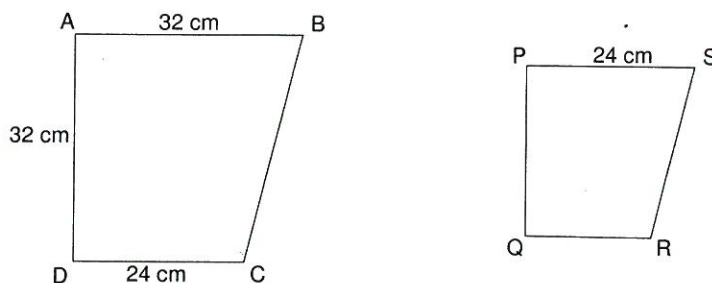


20. Consider quadrilaterals ABCD and MNOP shown below:
Given that quadrilateral ABCD $\sim$ quadrilateral NOPM, find



- (a) the size of angle NOP.
(b) the length of MN.
(c) the length of MP.
(d) the length of BC.
21. Answer true or false for each of the following statements:
- | | |
|--|--|
| (a) All rectangles are similar to each other. | (b) All squares are similar to each other. |
| (c) All rhombi are similar to each other. | (d) All circles are similar to each other. |
| (e) All parallelograms are similar to each other. | (f) All regular pentagons are similar to each other. |
| (g) All triangles are similar to each other. | (h) All isosceles triangles are similar to each other. |
| (i) All equilateral triangles are similar to each other. | (j) All right triangles are similar to each other. |
| (k) The angles of similar figures are equal. | (l) The sides of similar figures are in the same ratio. |
| (m) Triangles with two pairs of equal angles are similar. | (n) Similar figures must have the same shape and different size. |
| (o) Quadrilaterals with corresponding sides in the same ratio are similar. | (p) All rectangles with corresponding sides in the same ratio are similar. |

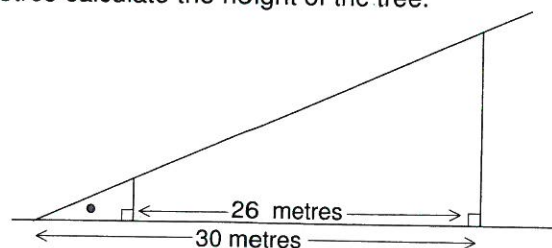
22. The quadrilaterals below are similar.



- (a) Complete the statement quadrilateral ABCD $\sim$ quadrilateral _____
(b) What scale factor has been used to reduce quadrilateral ABCD to its image?
(c) Find the length of side QR.
(d) Find the lengths of BC and SR giving your answers correct to 2 decimal places if appropriate.

EXERCISE 14D

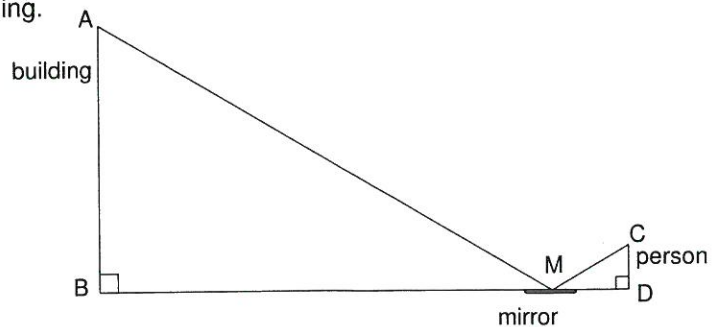
1. The height of a tree is to be found using a metre ruler. The ruler is placed vertically in the tree's shadow so that the ruler's shadow coincides with the end of the tree's shadow. If the ruler is 26 metres from the tree and the tree's shadow is 30 metres calculate the height of the tree.



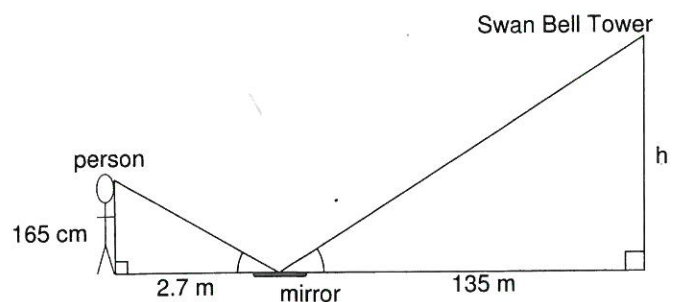
2. A 5 metre flagpole casts a shadow of length 12 metres at the same time as a nearby tower casts a shadow of 90 metres. Calculate the height of the tower.
3. At noon Frank who is 196 cm tall casts a shadow of 234 cm. At the same time a gum tree casts a shadow of length 31.5 metres. Calculate the height of the gum tree to the nearest tenth of a metre.
4. A mobile telephone tower casts a shadow of 36 metres. At the same time a 2 metre pole held vertically casts a 2.5 metre shadow. Find the height of the telephone tower.
5. A 1.5 metre pole is placed in the shadow of a tree so that ends of their shadows coincide.
 (a) Find the height of the tree if the tree's shadow is 18 m and the pole's 2 m.
 (b) Around midday it is observed that the tree's shadow is 8.1 m long. Find the length of the pole's shadow at this time.

6. Triangles ABC and PQR are similar and triangle ABC has sides of length 108 cm, 144 cm and 126 cm. The length of the shortest side of triangle PQR is 138 cm. Find the lengths of the other two sides of triangle PQR.

7. A mirror is placed on the ground so that the top of a building can be seen by a person looking into the mirror towards the building such that $\angle AMB = \angle CMD$. A person whose eyes are 164 cm above the ground is standing 3 metres from the mirror whereas the foot of the building is 75 metres from the mirror. Calculate the height of the building.



8. The height of the Swan Bell Tower which stands on the bank of the Swan River in Perth W.A. is to be found using a mirror. Use the information given in the diagram below to find the height of the tower.

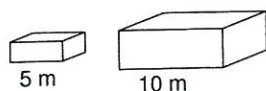


9. The foot of a ladder is 1.5 metres from a fence that is 2 metres high. The ladder touches the fence and rests on the window sill of a building that is 6.6 metres on the other side of the fence.
- (a) Calculate the height of the window sill above the ground. (b) How long is the ladder?

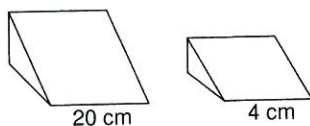
EXERCISE 14F

For questions 1 to 6 each pair of solids is similar. Find (a) the ratio of the surface area of the smaller solid to that of the larger, and (b) the ratio of the volume of the smaller solid to that of the larger.

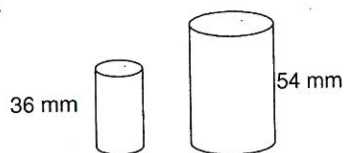
1.



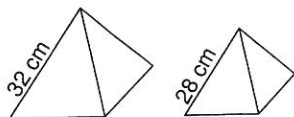
2.



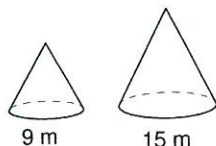
3.



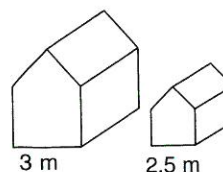
4.



5.



6.



7.

Two cubes have side lengths of 10 cm and 6 cm respectively.

(a) What important fact about cubes allows us to compare the ratios of the lengths of their corresponding sides?

(b) What scale factor must be applied to transform the larger cube so that its image is the smaller?

(c) Calculate the ratio of the areas of corresponding faces.

(d) Calculate the ratio of their volumes.

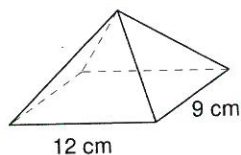
(e) The surface area of the larger cube is 600 cm^2 . **Show** how to use the scale factor found in (b) to calculate the surface area of the smaller cube.

(f) The volume of the smaller cube is 216 cm^3 . Using the scale factor **show** how to find the volume of the larger cube.

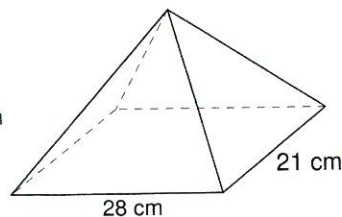
8.

A rectangular pyramid of base dimensions 12 cm by 9 cm is enlarged to give a pyramid with base dimensions 28 cm by 21 cm.

(a) What is the scale factor?

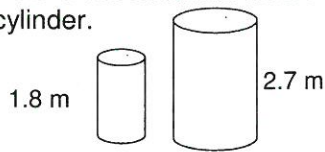


(b) If the volume of the smaller pyramid is 540 cm^3 , what is the volume of the larger?

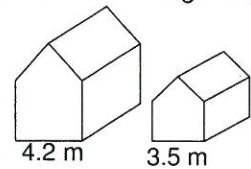


(c) If the surface area of the larger pyramid is 1176 cm^2 , what is the surface area of the smaller?

9. The cross-section of the larger cylinder is 3.6 m^2 . Given that the cylinders are similar calculate the cross-sectional area of the smaller cylinder.



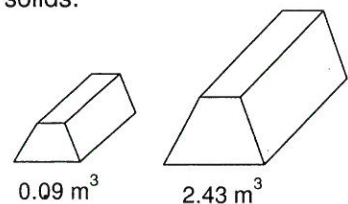
10. The two solids shown below are similar. If the volume of the smaller solid is 150 m^3 calculate the volume of the larger solid.



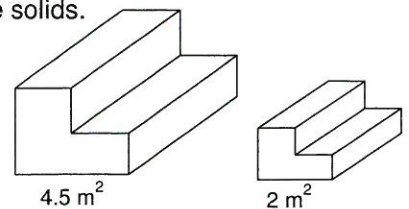
11. The area of the circular bases of two similar cones are 108 m^2 and 147 m^2 respectively.
 (a) Calculate the scale factor. (b) If the slant height of the smaller cone is 15 cm, what is the slant height of the larger cone?

- (c) If the volume of the smaller cone is 648 cm^3 , what is the volume of the larger cone? (d) What is the ratio of their surface areas?

12. The volumes of the two similar solids shown below are 0.09 m^3 and 2.43 m^3 respectively. Note that the diagrams are not drawn to scale.
 (a) Find the ratio of the lengths of the corresponding sides of these solids.

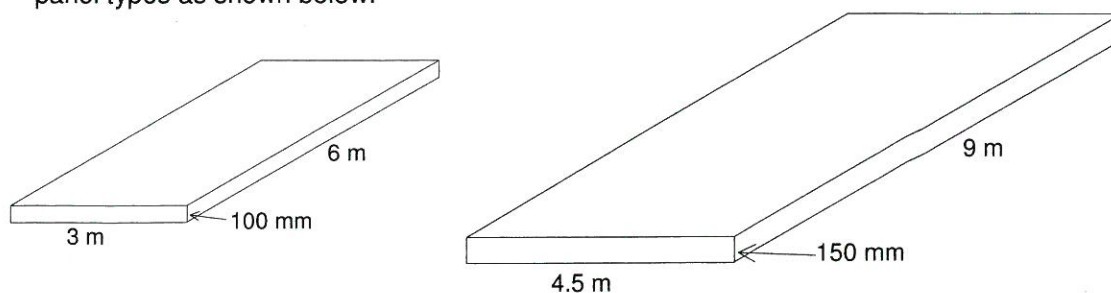


- (b) What is the ratio of their surface areas?
 (c) What is the ratio of the areas of their top faces?
 (d) If the width of the smaller solid is 40 cm find the width of the larger solid.
 (e) If 1.35 litres of paint is needed to paint the larger solid, how much paint is needed to paint the smaller solid?
13. The surface areas of the two similar solids shown below are 4.5 m^2 and 2 m^2 respectively. Note that the diagrams are not drawn to scale.
 (a) Find the ratio of the lengths of the corresponding sides of these solids.



- (b) What is the ratio of the areas of the 'L' shapes in these solids?
 (c) What is the ratio of the volumes of these solids?
 (d) If the height of the larger solid is 84 cm, what is the height of the smaller?
 (e) It costs \$135 to paint the larger solid, how much does it cost to paint the smaller solid?

14. A manufacturer of precast concrete panels used in the building industry manufactures two similar panel types as shown below.



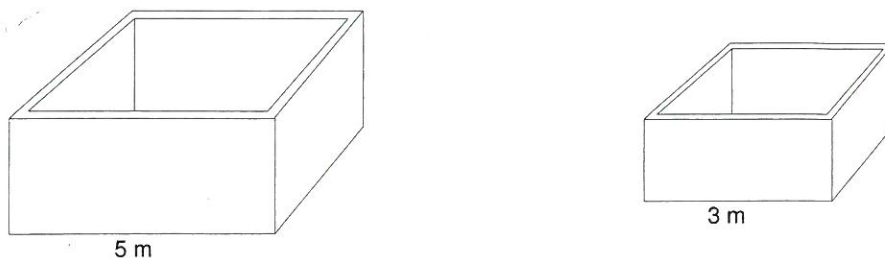
- (a) The cost of concrete to make the smaller panel is \$280. Calculate the cost of concrete to make the larger panel

Before being used the panels require one coat of a special sealant which costs \$35/L. One litre of this sealant covers an area of 3.5 square metres.

- (b) Calculate the cost of sealing the smaller panel.

- (c) How much will it cost to seal the larger panel?

15. Two similar water tanks are shown below:



- (a) It takes 25 hours to fill the larger tank with water. If filled at the same rate, how long will it take to fill the smaller tank?
- (b) If the internal surface area of the smaller tank is 72 square metres. What is the internal surface area of the larger tank?
- (c) The cost of concrete to make the larger tank is \$1250. Find the cost of concrete required to make the smaller tank.

16. If all cake recipe ingredients are doubled then the size of the resulting cake will be doubled. Comment.

9. (a) (i) 0.5, (ii) Triangle JKL ~ Triangle EDF 10. (a) (i) $\frac{4}{5}$, (ii) Square CDEF ~ Square JMLK 11. (b) Not similar, lengths of the corresponding sides are not in the same ratio
12. (b) Not similar, corresponding angles not equal and lengths of the corresponding sides are not in the same ratio.
13. (a) (i) 0.5, (ii) Kite ABCD ~ Kite SRUT 14. (a) (i) 0.75, (ii) Circle A ~ Circle B 15. (b) Not similar, corresponding angles not equal and lengths of the corresponding sides are not in the same ratio.
16. (a) $\frac{PR}{5} = \frac{8}{4}$, PR = 10 cm (b) BC = 3 cm
17. (a) Figure ABCDEF ~ Figure TSRQPU (b) $\frac{5}{8}$
(c) a = 1 m, b = 0.8 m, c = 0.3 m
18. (a) Yes because parallelogram AGFE is the image of parallelogram ABCD under the given enlargement transformation. (b) 1.5 (c) 7.8 cm
19. (a) $\frac{2}{3}$, (b) a = 6 cm, b = 12 cm, c = 10 cm, d = 15 cm
20. (a) 98° (b) 48 mm (c) 35.2 mm (d) 20 mm
21. (a) False (b) True (c) False (d) True (e) False (f) True (g) False (h) False (i) True (j) False (k) True (l) True (m) True (n) False (o) False (p) True 22. (a) Quad PSRQ (b) 0.75 (c) 18 cm (d) BC = 32.98 cm, SR = 24.74 cm

EXERCISE 14C

1. (a) (i) $\triangle ABC \sim \triangle HGJ$ (ii) AAA Similarity (iii) not possible.
(b) (i) $\triangle DEF \sim \triangle HGI$ (ii) SAS Similarity (iii) $\frac{7}{6}$
(c) (i) $\triangle ABC \sim \triangle GJH$ (ii) SSS Similarity (iii) $\frac{7}{8}$
(d) (i) $\triangle ABC \sim \triangle DEF$ or $\triangle FED$ (ii) SSS Similarity (iii) $\frac{9}{7}$
2. Triangles are similar. (a) $\triangle ABC \sim \triangle FED$ (b) AAA Similarity (c) 1.25 (d) a = 4 m, d = 7.5 m
3. Triangles are similar. (a) $\triangle ABCD \sim \triangle ROP$ (b) AAA Similarity (c) 0.8 (d) c = 10 cm, p = 12 cm
4. Triangles are similar. (a) $\triangle DEF \sim \triangle TRS$ (b) SAS Similarity (c) 2 (d) f = 3.875 m, x = 31°, y = 64°.
5. Triangles are similar. (a) $\triangle LMN \sim \triangle PRQ$ (b) SSS Similarity (c) $\frac{5}{6}$ (d) x = 83°, y = 41°
6. Triangles cannot be shown to be similar.
7. Triangles are similar. (a) $\triangle ABC \sim \triangle TSU$ (b) SAS Similarity (c) 0.8 (d) a = 14.375 m.
8. Triangles are similar. (a) $\triangle LMN \sim \triangle STR$ (b) RHS Similarity (c) 1.5 (d) m = 24 cm, t = 36 cm, x = 53°.
9. Triangles are similar. (a) $\triangle JKL \sim \triangle NML$ (b) AAA Similarity (c) 2.4 (d) m = 69.6 cm, n = 50.4 cm.
10. Triangles are similar. (a) $\triangle PTS \sim \triangle QTR$ (b) AAA Similarity (c) 0.5 (d) w = 61°, x = 14 mm, y = 6 mm.
11. Triangles cannot be shown to be similar.
12. (a) (i) 0.8 (ii) x = 32, y = 35 (b) (i) 1.5 (ii) x = 91, y = 40 (c) (i) 0.6 (ii) x = 36, y = 39 (d) (i) $1\frac{3}{4}$ (ii) x = 6, y = 7.5 (e) (i) 2 (ii) x = 12, y = 7 (f) (i) 0.4 (ii) x = 90, y = 112.5

EXERCISE 14D

1. 7.5 metres 2. 37.5 metres 3. 26.4 metres 4. 28.8 metres
5. (a) 13.5 metres (b) 0.9 metres 6. 161 cm and 184 cm
7. 41 metres 8. 82.5 metres 9. (a) 10.8 m (b) 13.5 m
10. (a) 48 metres (b) 189 m to nearest metre 11. 55 metres
12. (a) 45 cm (b) 60 cm

EXERCISE 14E

1. 1 : 4 2. 1 : 4 3. 25 : 36 4. 9 : 16 5. 64 : 225 6. 49 : 121
7. 7.2 cm² 8. 34.8 m² 9. 113.4 cm² 10. 1472 cm²
11. (a) (i) $\frac{2}{3}$ (ii) 8 cm (b) (i) 1.5 (ii) 24 m (c) (i) 0.75 (ii) 24 mm (d) (i) 0.8 (ii) 4 m 12. 7200 cm² 13. (a) 3 : 7 (b) 9 : 49
14. (a) 5 : 3 (b) 5 : 3 15. (a) (i) Perimeter is trebled (ii) Area is increased by a factor of 9. (b) (i) Perimeter is doubled (ii) Area is increased by a factor of 4. (c) (i) Circumference is increased by a factor of 4. (ii) Area is increased by a factor of 16. (d) (i) Perimeter is halved. (ii) New area is a quarter of the original area. (e) Area is doubled (triangles are not similar)

16. (a) 132 cm (b) 627 cm 17. (a) 532 cm (b) 12800 cm²
18. (a) $\frac{1}{5}$ (b) Area A is 25 times bigger than area B.
19. (a) 4 (b) Area A is $\frac{1}{16}$ th the area of B.
20. (a) 1.8 (b) Area A is $\frac{25}{81}$ of area B.
21. (a) $\frac{5}{7}$ (b) Area A is 1.96 time bigger than area B.
22. (a) 1.6 (b) Area A is $\frac{25}{64}$ of area B.
23. (a) $\frac{3}{5}$ (b) Area A is $2\frac{7}{9}$ times the area of B.
24. (a) 972 m² (b) 15 m
25. (a) radius = 5 cm, area 25π cm² (b) radius = 10 cm, area 100π cm² (c) 4 : 9

EXERCISE 14F

1. (a) 1 : 4 (b) 1 : 8 2. (a) 1 : 25 (b) 1 : 125
3. (a) 4 : 9 (b) 8 : 27 4. (a) 49 : 64 (b) 343 : 512
5. (a) 9 : 25 (b) 27 : 125 6. (a) 25 : 36 (b) 125 : 216
7. (a) All cubes are similar to each other. (b) 0.6 (c) 25 : 9 (d) 125 : 27 (e) Surface area = $0.6^2 \times$ surface area of large cube = $0.6^2 \times 600 = 216$ cm². (f) Volume = $(\frac{1}{0.6})^3 \times$ volume

of small cube = $(\frac{1}{0.6})^3 \times 216 = 1000$ cm³

8. (a) $2\frac{1}{3}$ (b) 6860 cm³ (c) 216 cm²
9. 1.6 m²
10. 259.2 m³
11. (a) $\frac{7}{6}$ (b) 17.5 cm (c) 1029 cm³ (d) 36:49
12. (a) 1 : 3 (b) 1 : 9 (c) 1 : 9 (d) 120 cm (e) 150 mL
13. (a) 3 : 2 (b) 9 : 4 (c) 27 : 8 (d) 56 cm (e) \$60
14. (a) \$945 (b) \$378 (c) \$850.50
15. (a) 5 hours 24 mins (b) 200 m² (c) \$270
16. Not true. To double the size of the cake all ingredients need to be 8 times the original ingredient quantities.
17. (a) 3200 cm² (b) 16 eggs (c) 8 serves
18. (a) 30 pounds (b) 6.38 inches (2 d.p.) (c) Surface area of a 6 inch cannonball is 4 times greater than that of a 3 inch cannonball.
19. (a) 186 cm (b) 1080 kg (c) 1 : 36
20. (a) 1 : 4 (b) 3200 cm³ (c) 22.9 mm (d) 1050 cm²

EXERCISE 14G

1. (a) (i) 1000 (ii) 1 cm : 10 m (b) (i) 10 000 (ii) 1 mm : 10 m (c) (i) 10 000 (ii) 1 m : 10 km (d) (i) 10 (ii) 1 cm : 0.1 m (e) (i) $\frac{1}{5}$ (ii) 1 m : 200 mm (f) (i) $\frac{1}{5}$ (ii) 1 km : 400 m (g) (i) $\frac{1}{4}$ (ii) 1 m : 250 mm (h) (i) $\frac{5}{4}$ (ii) 1 m : 125 cm
2. (a) 5000 : 1 (b) 1 : 3000 (c) 250 : 1 (d) 2000 : 1 (e) 20000 : 1 (f) 1 : 1200 (g) 1 : 80 (h) 2500 : 1 (i) 5000 : 1 (j) 1 : 300000 (k) 1 : 1500

Expect some small variation between your answers and the given answers in questions involving measuring lengths as measurements are estimates

EXERCISE 14H

1. 4 cm 2. 5 cm 3. 8 cm 4. 2.5 cm 5. 2.5 mm 6. 1.6 mm
7. 0.35 mm 8. 3.9 cm 9. 4 m 10. 700 m 11. 3.2 m 12. 14.75 km
13. 2.75 mm 14. 1.56 cm 15. 1.6 mm 16. 1.25 km
17. (a) 14.56 m (b) 16.8 m (c) 330 m (d) 5 mm (e) 90 m²
18. (a) 195 m (b) 1400 m² (c) 3.1 mm (d) 1.25 m

EXERCISE 14I

1. (a) 14 m (b) 6 m (c) 3.5 m (d) 8.75 m² (e) 63 m² (f) 12.25 m²
2. (a) 200 (b) 10.5 m (c) 10 m (d) 3.5 m (e) 4 m (f) 48 m (g) 14 m² (h) Area = $5.25 \times 5 \times 200$ cm² = 505000 cm² = 105 m²
3. (a) 4.4 m (b) 3.6 m (c) 1.2 m (d) 2.38 m² (e) 15.168 m²